

Figure 10.9 How to draw an ellipse.

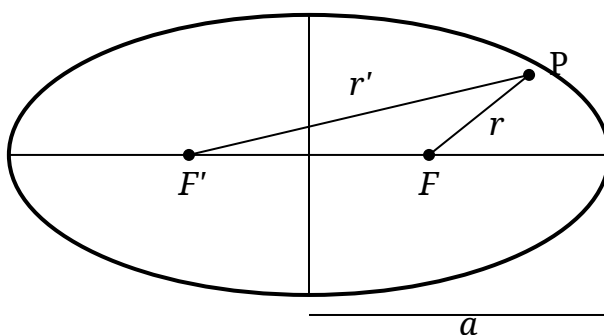


Figure 10.10 An ellipse. The total distance $r + r' + \overline{F'F}$ is a constant, so $r + r'$ must be constant. Points P' and P are called the apsides.

denoted a , and the semiminor axis is denoted b . The distance from the center of the ellipse to either focal point will be $f = ea$, where e is a number less than one, called the *eccentricity*. The eccentricity e and the semimajor axis a determine the size and shape of the ellipse.

It turns out that

$$r + r' = 2a.$$

This relation may surprise you, but it is easily proved. Consider point P. There $r = \overline{FP}$ and $r' = \overline{F'P}$. By symmetry, $\overline{F'P} = \overline{FP'}$ and therefore $r + r' = \overline{FP} + \overline{FP'} = 2a$. Since $r + r'$ is a constant, if it is equal to $2a$ at one point, it is equal to $2a$ at any point.

Now consider the triangle formed by r, r' , and $2f$, where $f = ea$. See [Figure 10.10](#). According to the law of cosines, if θ is the "external" angle,

$$r'^2 = r^2 + (2ae)^2 + 2r(2ae) \cos \theta. \quad (10.20)$$

Substituting $r' = 2a - r$ into [Equation \(10.20\)](#) gives

$$(2a - r)^2 = r^2 + 4a^2e^2 + 4aer \cos \theta,$$

and a little bit of algebra leads to

$$r = \frac{a(1-e^2)}{1+e \cos \theta}. \quad (10.21)$$

This is the equation for an ellipse in plane polar coordinates. If we were to go through the same process for hyperbolas and parabolas, we would obtain similar equations for $r = r(\theta)$. In fact, the equation for any conic section can be expressed in the form

$$r = \frac{p}{1+e \cos \theta}, \quad (10.22)$$

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where $p = a(1 - e^2)$ for an ellipse, $p = a(e^2 - 1)$ for a hyperbola,⁹ and $p = a$ for a parabola¹⁰ (a is a characteristic constant for each type of curve and, in general, $p = l^2/GMm^2$). The various curves correspond to different choices of the eccentricity e . Specifically,

- $e = 0$ gives a circle,
- $e < 1$ gives an ellipse,
- $e = 1$ gives a parabola,
- $e > 1$ gives a hyperbola.

The semimajor axis a determines the *size* of an ellipse and the eccentricity determines its *shape*. At zero eccentricity, the ellipse degenerates into a circle. As the eccentricity gets closer and closer to one, the ellipse gets flatter and flatter. That is, as e approaches unity, the ratio of the semiminor axis to the semimajor axis approaches zero. As you will prove in [Exercise 10.12](#),

$$\frac{b}{a} = (1 - e^2)^{1/2}. \quad (10.23)$$

You probably know from geometry that the area of an ellipse is given by $S = \pi ab$. It is sometimes convenient to write this in terms of a only. Using [Equation \(10.23\)](#),

$$S = \pi a^2(1 - e^2)^{1/2}. \quad (10.24)$$

Having considered the properties of conics, let's go back to the motion of planets. Comparing the equation of an ellipse (10.21) with the equation of the orbit (10.17), you can see that the orbit is

an ellipse (or, more generally, a conic section). A term-by-term comparison of these two equations shows that

$$\frac{l^2}{GMm^2} = a(1 - e^2), \quad (10.25)$$

and that

$$\frac{Al^2}{GMm^2} = e.$$

But we showed in Equation (10.19) that

$$A = \left[\frac{(GMm)^2 m^2}{l^4} + \frac{2Em}{l^2} \right]^{\frac{1}{2}},$$

so

$$e^2 = \frac{(GMm)^2 m^2}{l^4} \left(\frac{l^4}{(GMm)^2 m^2} + \frac{2Em}{l^2} \frac{l^4}{(GMm)^2 m^2} \right).$$

Therefore,

$$e^2 = 1 + \frac{2El^2}{m(GMm)^2}. \quad (10.26)$$

⁹ A hyperbola has two branches and is described by the equation

$$r = \frac{a(e^2 - 1)}{\pm 1 + e \cos \theta},$$

where the "plus branch" corresponds to an attractive force and the "minus branch" corresponds to a repulsive force. The quantity a is one half the distance between the vertices.

¹⁰ For a parabola $a = 2r_p$, where r_p is the distance from the focus to the nearest point on the parabolic curve.

This gives us an expression for the eccentricity in terms of constants of the motion. Note that for $E > 0$ we get $e > 1$, a hyperbola. For $E = 0$ we get $e = 1$, a parabola. For $E < 0$, we get $e < 1$, an ellipse. Finally, for a circle, $e = 0$ so $2El^2/m(GMm)^2 = -1$.

Plugging Equation (10.26) into (10.25) gives the following expression for the semimajor axis:

$$a = -\frac{GMm}{2E}. \quad (10.27)$$

The minus sign is necessary because the total energy is negative for an ellipse. Equation (10.27) shows that the semimajor axis depends only on the energy.

Worked Example 10.3

A comet of mass m starts from infinity with velocity v_0 and impact parameter b . (If undeflected, the path of the comet would be a straight line passing the Sun at a distance b .)

(a) Show that the distance of closest approach to the Sun is approximately $b^2 v_0^2 / 2GM$. (b) Write the equation of the orbit in polar coordinates in terms of M , v_0 , and b , where M is the mass of the Sun.

Solution (a) Let the Sun–comet distance at perihelion be denoted d . At that point, the speed of the comet is v_d and its velocity is perpendicular to the Sun–comet line.

By conservation of angular momentum $l_\infty = l_d$, and

$$mv_0 b = mv_d d \Rightarrow \frac{b}{d} = \frac{v_d}{v_0}.$$

By conservation of energy

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_d^2 - \frac{GMm}{d},$$

so dividing by $mv_0^2/2$,

$$1 = \left(\frac{b}{d}\right)^2 - \frac{2GM}{dv_0^2}.$$

Multiply by d^2 and obtain the quadratic

$$d^2 + \frac{2GM}{v_0^2}d - b^2 = 0,$$

$$d = -\frac{GM}{v_0^2} \pm \sqrt{\left(\frac{GM}{v_0^2}\right)^2 + b^2}.$$

Since $d > 0$, we use the positive sign and write

$$d = \frac{GM}{v_0^2} \left(-1 + \left[1 + \frac{b^2 v_0^4}{G^2 M^2} \right]^{1/2} \right).$$

Apply the binomial expansion to get

$$d = \frac{GM}{v_0^2} \left(-1 + \left[1 + \frac{1}{2} \frac{b^2 v_0^4}{G^2 M^2} + \dots \right] \right),$$

$$d \simeq \frac{GM}{v_0^2} \frac{1}{2} \frac{b^2 v_0^4}{G^2 M^2} = \frac{b^2 v_0^2}{2GM}.$$

(b) In polar coordinates the hyperbolic orbit under an attractive force is

$$r = \frac{a(e^2-1)}{1+e \cos \theta}.$$

The problem requires us to express a and e in terms of the given parameters, namely v_0 , b , and constants. Taking the derivative of r with respect to time we have

$$\dot{r} = \frac{a(e^2-1)}{(1+e \cos \theta)^2} e \sin \theta \dot{\theta},$$

Now $l = mr^2\dot{\theta}$ so

$$\dot{\theta} = \frac{l}{mr^2} = \frac{l}{m} \frac{(1+e \cos \theta)^2}{a^2(e^2-1)^2},$$

and we can write

$$\dot{r}^2 = \frac{e^2 l^2 / m^2}{[a(e^2-1)]^2} \sin^2 \theta.$$

The square of the speed of the comet in polar coordinates is

$$v^2 = \dot{r}^2 + r^2 \dot{\theta}^2.$$

The second term in v^2 is $r^2 \dot{\theta}^2$. But,

$$r \dot{\theta} = \frac{1}{r} \frac{l}{m} = \frac{l}{m} \frac{1+e \cos \theta}{a(e^2-1)},$$

$$\therefore r^2 \dot{\theta}^2 = \frac{l^2}{m^2} \frac{1+2e \cos \theta + e^2 \cos^2 \theta}{[a(e^2-1)]^2},$$

so

$$v^2 = \frac{l^2(1+2e \cos \theta + e^2(\sin^2 \theta + \cos^2 \theta))}{m^2[a(e^2-1)]^2}$$

$$\begin{aligned}
&= \frac{l^2}{m^2 a(e^2 - 1)} \left(\frac{2 + 2e \cos \theta + e^2 - 1}{a(e^2 - 1)} \right) \\
&= \frac{l^2}{m^2 a(e^2 - 1)} \left[\frac{2}{r} + \frac{1}{a} \right].
\end{aligned}$$

Comparing $r = \frac{a(e^2 - 1)}{1 + e \cos \theta}$ for a hyperbolic orbit with the general form of the equation of the orbit, [Equation \(10.17\)](#), we appreciate that

$$\frac{l^2}{GMm^2} = a(e^2 - 1).$$

Consequently,

$$v^2 = \frac{GMm^2 a(e^2 - 1)}{m^2 a(e^2 - 1)} \left[\frac{2}{r} + \frac{1}{a} \right] = GM \left[\frac{2}{r} + \frac{1}{a} \right].$$

At $r = \infty$, the velocity is v_0 and we see that

$$v_0^2 = \frac{GM}{a},$$

so

$$a = \frac{GM}{v_0^2}.$$

Also

$$a(e^2 - 1) = \frac{l^2}{GMm^2}.$$

Solving for e we obtain

$$e = \left(1 + \left[\frac{v_0^2 b}{(GM)} \right]^2 \right)^{1/2},$$

and finally

$$r = \frac{v_0^2 b^2 / GM}{1 + \sqrt{1 + \left(\frac{bv_0^2}{GM}\right)^2} \cos \theta}.$$

Exercise 10.11

Starting with Equation (10.20), obtain Equation (10.21). ■

Exercise 10.12

Show that for an ellipse $b/a = (1 - e^2)^{1/2}$. (Hint: Apply the Pythagorean theorem to the triangle FOP in Figure 10.11.) ■

Exercise 10.13

A planet is in an elliptical orbit with semimajor axis a . By averaging the largest and smallest values of r , show that it leads to an "average" value of the potential energy equal to $-GMm/a$. (Compare with the virial theorem discussed in Worked Example 10.2.) ■

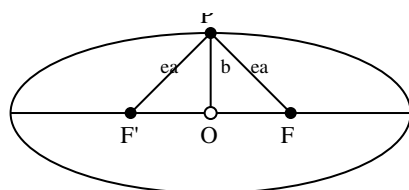


Figure 10.11 An ellipse. Note that the distance from the center to the focal point is ea and that the sum of the distances FP and $F'P$ is $2a$.

Exercise 10.14

Plot Equation (10.22) for $e = 2$, $e = 1.0$, $e = 0.5$, and $e = 0$. ■

10.9 Kepler's Laws Revisited

Having applied Newton's second law and Newton's law of universal gravitation to the problem of

planetary motion, let us now return to Kepler's laws that were given in [Section 10.2](#). You will find that Kepler's laws are simply descriptions of some aspects of the theory we have developed.

Recall that Kepler's first law states:

The orbit of a planet is an ellipse with the Sun at one of the focal points.

We applied Newton's second law to a particle of mass m in the gravitational field of M , writing $m\ddot{\mathbf{r}} = (-GMm/r^2)\hat{\mathbf{r}}$. After a significant amount of math, this led to the equation of the orbit $r = r(\theta)$, which we recognized as the equation of an ellipse. Thus Kepler's first law is a consequence of Newton's law of universal gravitation. Actually, the proof is even more general than Kepler's first law, for we have seen that objects in the solar system acting under the inverse square gravitational force will move in a trajectory which may be an ellipse, but it could also be one of the other conic sections.

Kepler studied the orbit of Mars all his life, and he deduced other important facts about its motion. For one thing, he found that the motion of Mars as a function of time was not uniform. Mars speeds up as it approaches the Sun, moves fastest at its point of nearest approach, then slows down as it moves away from the Sun. He expressed this fact in a quantitative way in his "Law of Areas" or, as we call it, Kepler's second law:

The radius vector of a planet sweeps out equal areas in equal times.

This is illustrated in [Figure 10.12](#). If the planet moves from a to b in the same amount of time as to go from c to d then, according to Kepler's second law, the two shaded areas are equal. So another way to state Kepler's second law is, "The *areal velocity* of a planet is constant."

In point of fact, we already proved Kepler's second law, but you might not have noticed it because it was stated in quite a different way. We said, "For a central force, the angular momentum is constant." Are the two statements equivalent? Does the fact that the angular momentum is constant imply that the *areal velocity* is constant? The answer is yes. To prove that this is true, consider a planet located at vector position $\mathbf{r}$. After a time interval dt the planet will be at $\mathbf{r} + d\mathbf{r} = \mathbf{r} + \mathbf{v}dt$. See [Figure 10.13](#). The shaded region is the area swept out. Recall that $|\mathbf{a} \times \mathbf{b}|$ is equal to the area of a parallelogram whose sides are the vectors $\mathbf{a}$ and $\mathbf{b}$. The shaded region in [Figure 10.13](#) is one half of the parallelogram formed from $\mathbf{r}$ and $d\mathbf{r}$. Therefore, the area swept out by the planet's radius vector in time dt is

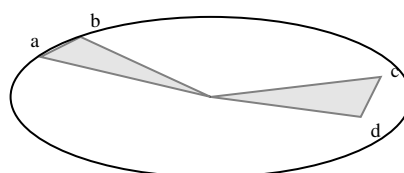


Figure 10.12 Kepler's second law states that equal areas are swept out in equal times.

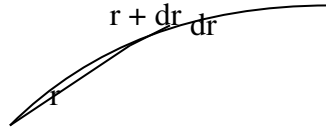


Figure 10.13 The area of the triangle is one half the area of the parallelogram. Therefore, $\Delta S = (1/2)|\mathbf{r} \times d\mathbf{r}| = (1/2)|\mathbf{r} \times \mathbf{v}\Delta t|$.

$$dS = \frac{1}{2}|\mathbf{r} \times \mathbf{v}dt| = \frac{dt}{2}|\mathbf{r} \times \mathbf{v}|.$$

But by the definition of angular momentum, $|\mathbf{r} \times \mathbf{v}| = l/m$. Therefore,

$$dS = \frac{1}{2} \frac{l}{m} dt,$$

or

$$\frac{dS}{dt} = \text{areal velocity} = \frac{l}{2m}.$$

The mass of a planet is constant, so the areal velocity is proportional to the angular momentum and $\frac{dS}{dt} = \text{constant}$ is completely equivalent to the relation $l = \text{constant}$. Kepler's second law is simply a consequence of the fact that in central force motion, the angular momentum is conserved.

Kepler's third law states:

The period of a planet squared is proportional to its semimajor axis cubed.

In mathematical terms, if τ is the time required for a planet to orbit the Sun and if a is the semimajor axis of the orbit, the third law can be written

$$\tau^2 = Ka^3,$$

where K is some constant. Kepler discovered this law after years of studying planetary data. On the other hand, our analytical study of elliptical motion leads to it in a fairly simple way. We just note that

$$\frac{dS}{dt} = \frac{l}{2m} \Rightarrow \oint_{\text{orbit}} dS = \frac{l}{2m} \oint_{\text{orbit}} dt.$$

Let S equal the area of the ellipse. Then if τ is the period,

$$S = \frac{l}{2m}\tau.$$

But we have seen that $S = \pi ab = \pi a^2(1 - e^2)^{1/2}$ so

$$\frac{l}{2m}\tau = \pi a^2(1 - e^2)^{1/2}.$$

According to Equation (10.25), for an ellipse, $(1 - e^2) = l^2/GMm^2a$. Squaring we obtain

$$\frac{l^2}{4m^2}\tau^2 = \pi^2 a^4 \frac{l^2}{GMm^2a},$$

or

$$\tau^2 = \frac{4\pi^2}{GM}a^3.$$

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This relation states that the period squared is proportional to the semimajor axis cubed. Thus, Kepler's third law is proved. Note that the constant of proportionality depends only on the mass of the Sun, so the ratio τ^2/a^3 should be the same for all the planets. Therefore, determining the period of a planet or asteroid allows us to evaluate its semimajor axis. Actually, this result is not completely accurate. There is a very small correction that is due to the fact that the Sun is not really at rest at the origin. The Sun and the planet move around their common center of mass. This point is almost, but not quite, at the center of the Sun. This introduces a small correction into Kepler's third law so that the quantity M in the denominator should be replaced by $M + m$. Once again we appreciate the technique in physics of first solving an easier idealized problem and introducing the complications later. (Problem 10.11 asks you to carry out the appropriate calculations.)

Worked Example 10.4

A particle moves in an elliptical orbit with semimajor axis a and eccentricity e . The velocity of the particle is observed to vary from a minimum value v_1 to a maximum value v_2 . Determine the period.

Solution The angular momentum of the particle is constant:

$$l = mvr = \text{constant}.$$

Therefore, the maximum (and minimum) speeds occur when r is at the smallest (and greatest) distance from the force center. Call these distances r_1 and r_2 . By conservation of angular momentum,

$$mv_1r_1 = mv_2r_2.$$

The points at r_1 and r_2 as well as the force center all lie on the semimajor axis, and

$$r_1 + r_2 = 2a.$$

Hence

$$\begin{aligned} l &= mv_1(2a - r_2) = mv_2r_2, \\ \therefore r_2 &= \frac{2av_1}{v_1 + v_2}, \end{aligned}$$

so the angular momentum can be written

$$l = m \frac{2av_1v_2}{v_1 + v_2}.$$

The angular momentum is related to the areal velocity by

$$\frac{l}{2m} = \frac{dS}{dt} = \text{areal velocity}.$$

The period is equal to the area divided by the areal velocity

$$\tau = \frac{\text{area}}{\text{areal velocity}} = \frac{\pi ab}{(l/2m)} = \frac{\pi ab(2m)}{m \frac{2av_1v_2}{v_1 + v_2}} = \frac{\pi b(v_1 + v_2)}{v_1v_2}.$$

By Exercise 10.12 the semiminor axis, b , is equal to $a(1 - e^2)^{1/2}$, so

$$\tau = \pi a(1 - e^2)^{1/2} \frac{v_1 + v_2}{v_1v_2}.$$

Worked Example 10.5

Suppose you were familiar with Kepler's work, so you knew that the orbits of planets are ellipses and that the angular momentum is constant. This suggests to you that the force on the planet is a central force, so $F = f(r)$, but you don't know any other properties of the force. Show that the force obeys the inverse square law, that is, show that $f(r) \propto 1/r^2$.

Solution In polar coordinates the radial equation of motion is

$$\ddot{r} - r\dot{\theta}^2 = -f(r)/m.$$

The angular equation of motion leads to

$$\dot{\theta} = l/mr^2.$$

Substituting,

$$\ddot{r} - \frac{l^2}{m^2 r^3} = -\frac{f(r)}{m}.$$

In terms of $u = 1/r$, recalling that $\ddot{r} = -\frac{l^2 u^2}{m^2} \frac{d^2 u}{d\theta^2}$, we obtain

$$\begin{aligned} -\frac{l^2 u^2}{m^2} \frac{d^2 u}{d\theta^2} - \frac{l^2 u^3}{m^2} &= -\frac{f(r)}{m}, \\ \frac{d^2 u}{d\theta^2} + u &= \frac{m}{l^2 u^2} f(r). \end{aligned} \tag{10.28}$$

Now the equation of an ellipse has the form

$$\begin{aligned} r &= \frac{p}{1+e \cos \theta}, \\ \therefore u &= \frac{1}{p} + \frac{e}{p} \cos \theta. \end{aligned}$$

Taking the derivative with respect to θ twice yields

$$\frac{d^2 u}{d\theta^2} = -\frac{e}{p} \cos \theta,$$

and Equation (10.28) becomes

$$\begin{aligned} -\frac{e}{p} \cos \theta + \left(\frac{1}{p} + \frac{e}{p} \cos \theta \right) &= \frac{m}{l^2 u^2} f(r) \\ \frac{1}{p} &= \frac{m}{l^2 u^2} f(r) = \frac{m}{l^2} r^2 f(r). \end{aligned}$$

That is,

$$f(r) = \frac{l^2}{mp} \frac{1}{r^2} \propto \frac{1}{r^2}.$$

Q.E.D.

Exercise 10.15

A certain asteroid has a period of four years. What is its semimajor axis in AU? (Assume you do not know the gravitational constant G or the mass of the Sun, but you do know that the Earth is at a distance of 1 AU from the Sun.) Answer: 2.52 AU. ■

Exercise 10.16

(a) Using the form $\tau^2 = (4\pi^2/GM)a^3$ and assuming you do not know the mass of the Sun, determine the semimajor axis of Saturn in AU, given its period is 29.5 Earth years. (b) Using the correction $(M + m)$ and looking up the appropriate values, obtain a corrected value. Answers: 9.55 AU, 9.52 AU. ■

Exercise 10.17

For a planet in a circular orbit, $F = ma$ can be written as $GMm/r^2 = mv^2/r$. Use this to derive Kepler's third law for circular orbits. ■

Exercise 10.18

Halley's comet has an eccentricity of 0.967. Its perihelion distance is 8.81×10^{10} m. What is its period? Answer: 75.4 years. ■

10.10 Orbital Mechanics

The first artificial satellite (Sputnik 1) was launched by the USSR on October 4th, 1957. Since then, nearly 10,000 satellites have been launched and about half of them are still in orbit. Artificial earth satellites have become an integral part of modern life. The Global Positioning System (GPS) is a constellation of satellites that can specify a location on the Earth to within an accuracy of up to 30 cm. You probably use it nearly every day. Satellites are used for navigation, weather forecasting, communications, oceanography, atmospheric science, astronomy, and

military purposes.

This section is a brief consideration of some of the physics used in analyzing the motion of artificial satellites. The two principal physical concepts are energy and angular momentum.

10.10.1 Energy and Angular Momentum of a Satellite

The "specific energy" of a satellite is usually expressed as

$$\mathcal{E} = \frac{1}{2}v^2 - \frac{\mu}{r},$$

where $\mathcal{E} = E/m$ is the energy per unit mass of the satellite and $\mu = GM$ is an abbreviation commonly used in celestial mechanics.¹¹ M is the mass of the Earth.

The angular momentum per unit mass of an object is denoted $\mathbf{h}$. By definition

$$\mathbf{h} = \mathbf{l}/m = \mathbf{r} \times \mathbf{v}.$$

and is a constant of the motion.

Artificial satellites are placed in elliptical or circular orbits. The most important parameters of such orbits are the eccentricity (e) and the semimajor axis (a). Two other useful parameters are the distances from the center of the Earth to perigee (r_p) and to apogee (r_a). At these two points the velocity vector $\mathbf{v}$ and the position vector $\mathbf{r}$ are perpendicular to one another and we can write

$$h = r_a v_a = r_p v_p.$$

¹¹ I will use μ for GM in this section only.